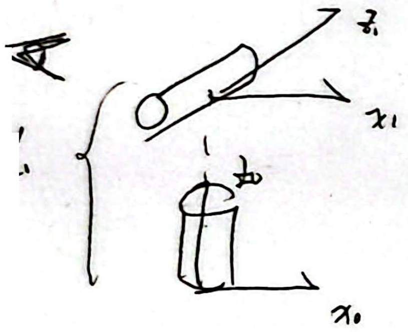
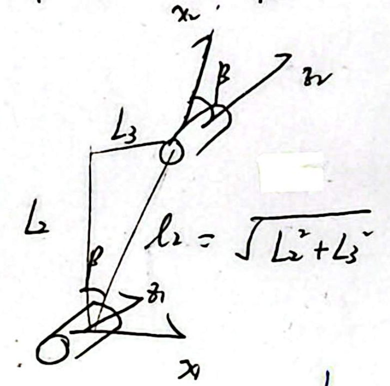
(4.11) (D-H)

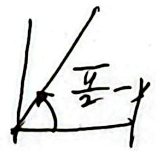
Examined (✓)

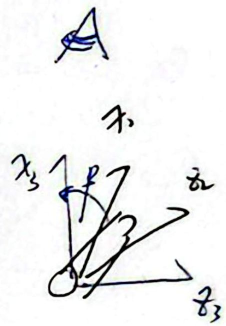
	沿 x_0: a	沿 x_0: α	沿 z_{0-1}: d	沿 z_{0-1}: θ
0 – 1	0	$\frac{\pi}{2}$	L_1	$0+\varphi_1$
1 – 2	$\sqrt{L_1^2+L_2^2}$	0	0	$(\frac{\pi}{2}-\beta)+\varphi_2$
2 – 3	0	$\frac{\pi}{2}$	0	$\beta+\varphi_3$
3 – 4	0	0	L_4+L_5	φ_4

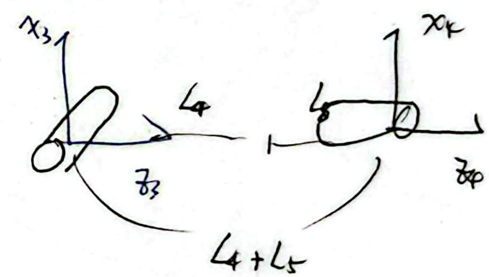




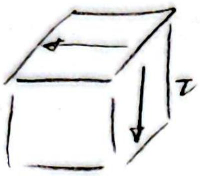
$$\beta = \arctan\left(\frac{L_3}{L_2}\right)$$

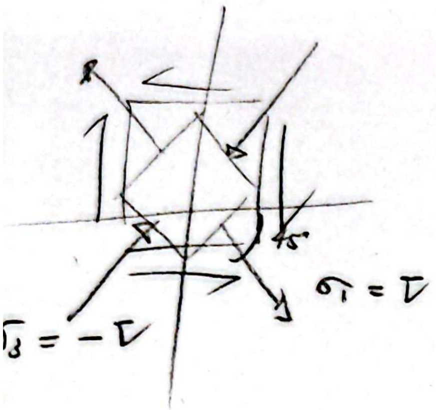






8. 纯剪切：无体积改变，仅形状改变



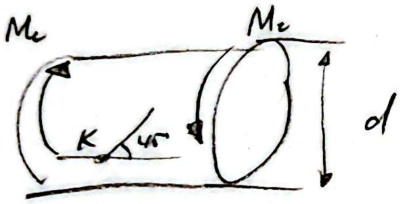


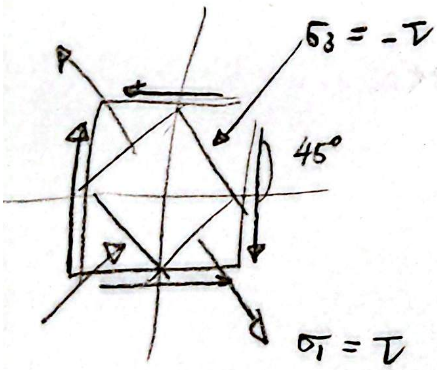
$\sigma_1 = \tau,\quad \sigma_2 = 0,\quad \sigma_3 = -\tau$

$$\theta = \frac{1-2\nu}{E}(\sigma_x+\sigma_y+\sigma_z)$$

$$= \frac{1-2\nu}{E}(\underset{\tau}{\sigma_1}+\underset{0}{\sigma_2}+\underset{-\tau}{\sigma_3})$$

$$= 0 \ \#$$





$$\varepsilon_{45^\circ} = \frac{1}{E}(\sigma_3 - \nu\sigma_1)$$

$$= \frac{1}{E}(-\tau-\nu\tau) = \frac{-(1+\nu)\tau}{E}$$

$$\therefore\ \tau = \frac{-E\varepsilon_{45^\circ}}{1+\nu}$$

$$\tau = \frac{M_e}{W_t},\quad M_e = \tau W_t = -\frac{E\varepsilon_{45^\circ}}{1+\nu}\cdot\frac{\pi d^3}{16}$$

应力偏张量 ⇒ 广义胡克定律

$tr(A) = \sum_{i=1}^{n} A_{ii} = A_{11} + A_{22} + \cdots A_{nn}$

$\sigma_{kk} = \sigma_{11} + \sigma_{22} + \sigma_{33}$

$\varepsilon_{kk} = \frac{\Delta V}{V} = \theta$

性质 $tr(A+B) = tr(A) + tr(B)$

$tr(A) = tr(A^T)$

回顾: $\sigma_{ij} = \underbrace{S_{ij}}_{\text{应力偏张量}} + \sigma_m \delta_{ij}$ — Kronecker

平均应力 $(\sigma_m) = \frac{1}{3}\sigma_{kk}$

$\varepsilon_{ij} = \underbrace{e_{ij}}_{\text{应变偏张量}} + \varepsilon_m \delta_{ij}$

— $\varepsilon_m = \frac{1}{3}\varepsilon_{kk} = (\theta)$ 体应变

$$\varepsilon_{ij} = \frac{1+\nu}{E}\sigma_{ij} - \frac{\nu}{E}\sigma_{kk}\delta_{ij}$$

$$\Leftrightarrow \quad \sigma_{ij} = 2G\varepsilon_{ij} + \lambda\varepsilon_{kk}\delta_{ij}$$

Step① 推导 求迹 (Trace) (= 令 $i=j$ 并求和)

$\sigma_{kk} = 3\sigma_m$

胡克 $\sigma_{ij} = 2G\varepsilon_{ij} + \lambda\varepsilon_{kk}\delta_{ij}$ 求迹

$$\sigma_{kk} = 2G\varepsilon_{kk} + 3\lambda\varepsilon_{kk} = (2G + 3\lambda)\varepsilon_{kk}$$

(体积模量) $K = \lambda + \frac{2}{3}G$

$$\sigma_{kk} = 3K\varepsilon_{kk} \Rightarrow 3\sigma_m = 3K(3\varepsilon_m) \Rightarrow \sigma_m = 3K\varepsilon_m$$

Step② 应力偏张量关系

$$S_{ij} = \sigma_{ij} - \sigma_m\delta_{ij}$$

广义胡克: $\sigma_{ij} = 2G\varepsilon_{ij} + \lambda\varepsilon_{kk}\delta_{ij}$

$$\therefore S_{ij} = 2G\left(\varepsilon_{ij} - \frac{1}{3}\varepsilon_{kk}\delta_{ij}\right)$$

$$\because e_{ij} = \varepsilon_{ij} - \varepsilon_m\delta_{ij}$$

$\therefore S_{ij} = 2Ge_{ij}$ (形状改变部分)

$\sigma_m = 3K\varepsilon_m$ (体积 ……)

Mises 屈服准则（形状改变比能准则）

观点：材料屈服仅由形状改变比能（Distortion Energy Density）达临界值引起，与体积改变比能（静水压力）无关

（弹性应变比能）$u = \frac{1}{2}\sigma_{ij}\varepsilon_{ij}$

应力 $\sigma_{ij} = S_{ij} + \sigma_m \delta_{ij}$

应变 $\varepsilon_{ij} = e_{ij} + \varepsilon_m \delta_{ij}$

$$u = \frac{1}{2}(S_{ij} + \sigma_m\delta_{ij})(e_{ij} + \varepsilon_m\delta_{ij})$$

$$= \frac{1}{2}(S_{ij}e_{ij} + S_{ij}\varepsilon_m\delta_{ij} + \sigma_m\delta_{ij}e_{ij} + \sigma_m\delta_{ij}\varepsilon_m\delta_{ij})$$

$\because$ $S_{ij}\delta_{ij} = S_{ii} = 0$ （tr（偏张量）= 0）

$$\therefore \quad u = \frac{1}{2}S_{ij}e_{ij} + \frac{3}{2}\sigma_m\varepsilon_m = u_d + u_v$$

形状改变比能　　体积改变比能

Step ②

偏张量胡克定律 $S_{ij} = 2Ge_{ij}$

$$u_d = \frac{1}{2}S_{ij}\left(\frac{S_{ij}}{2G}\right) = \frac{1}{4G}S_{ij}S_{ij}$$

展开 $S_{ij}S_{ij}$ （= 应力偏张量不变量 J_2 两倍）

$$u_d = \frac{1}{12G}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right]$$

Mises 临界值（单向拉伸实验）

$\sigma_1 = \sigma_s$（屈服强度），$\sigma_2 = \sigma_3 = 0$

$$u_{d,cr} = \frac{1}{12G}\left[(\sigma_s-0)^2 + (0-0)^2 + (0-\sigma_s)^2\right] = \frac{\sigma_s^2}{6G}$$

故：屈服条件

$$\sigma_s = \frac{1}{\sqrt{2}}\sqrt{(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2}$$